Lodestar: an explainable cognitive growth agent

AI4Brain, CISTAC 1.0 — technical report

AI4Brain team

October 2026

Abstract

Lodestar is an AI agent that assesses sixteen cognitive skills through short games, keeps a dynamic, historised profile of each person (the *Brain Twin*) in which every estimate carries an 80% range, coaches in natural language with every number traced to a cited fact, and adapts exercises, goals and plans as the profile changes. A LangGraph agent with 22 nodes decides what to do next and records every decision with the options it weighed. A Kalman-Elo scoring engine turns answers into estimates with honest uncertainty; four learned or statistical models forecast growth, detect drifting engagement, find the fatigue point and pick the game that helps most. On 200 simulated people with known skills, the profile recovers the true skill with correlation 0.82 after three sessions and 0.91 after ten, and its 80% ranges contain the truth 79.9% of the time. The system runs without camera or microphone, without a cloud model, and degrades gracefully when its models or memory are down. Lodestar is a practice coach; it never assesses medical or psychological conditions.

1 Objectives and scope

The brief asks for an agent that *assesses* the user across cognitive dimensions non-clinically, *understands* performance as an uncertainty-aware percentage profile, *coaches* with explainable, personalised feedback, and *improves* through adaptive exercises and measurable learning paths (Layers A–C), with an AI Collaboration Score, documented ethics, and optional voice, emotion-aware and predictive features. Table 1 maps each requirement to the component that meets it. All three bonus features are implemented.

Requirement	Lodestar
A: tasks per dimension; score, RT, errors, behaviour	43 browser games over 16 skills; every trial stores correctness, response time, hint level, practice flag and optional confidence
A: each dimension documented	Generated skill pages: definition, tasks, scoring, uncertainty, biases and evidence level
A: non-biometric mode mandatory	The default; camera and voice tone are off, per-switch consented, and never write to the profile
Brain Twin: history; projections with uncertainty	Append-only event log, daily history, growth forecast as an 80% range
B: explainable coaching, no clinical terms	Facts packet → draft → verify → repair; citations on every number; banned-word check in lint and at runtime
C: adaptation, plans, radar, curves, heatmaps, goals	Difficulty targeting, bandit, goals, personal experiments; all four chart types with ranges
AI Collaboration Score (0–100, formula)	Five components measured in play; Bayesian shares; documented weights and sensitivity
Ethics & data	Layered consent, export, deletion with receipt, retention job, encrypted user keys
Bonus	Voice in/out with wake word; on-device strain cues; predictive growth

Table 1: Brief requirements and where they are met.

2 Related work and positioning

Commercial brain-training apps (Lumosity, CogniFit, Elevate) report scores and streaks, but not how sure they are, why a game was chosen, or what the score means for the person. Adaptive practice systems in education solved the measurement half: Math Garden updated player and item ratings after every answer with an Elo system and a high-speed high-stakes scoring rule for over 150,000 children [1], and recent dynamic-K Elo variants let uncertainty grow when practice stops. Item response theory and computerised adaptive testing give the statistical frame, but are built for one-off tests rather than a profile that keeps changing. On the coaching side, language models write

fluent feedback but invent numbers, and AI assistance can make people over-rely on it [2]. Lodestar combines the three: a Kalman/Glicko-style Elo per skill with honest ranges, a graph agent whose decisions are code and are recorded, and a coach that may only say numbers it can cite. It also turns the over-reliance problem into something to train and measure, with the AI Collaboration Score.

3 Architecture

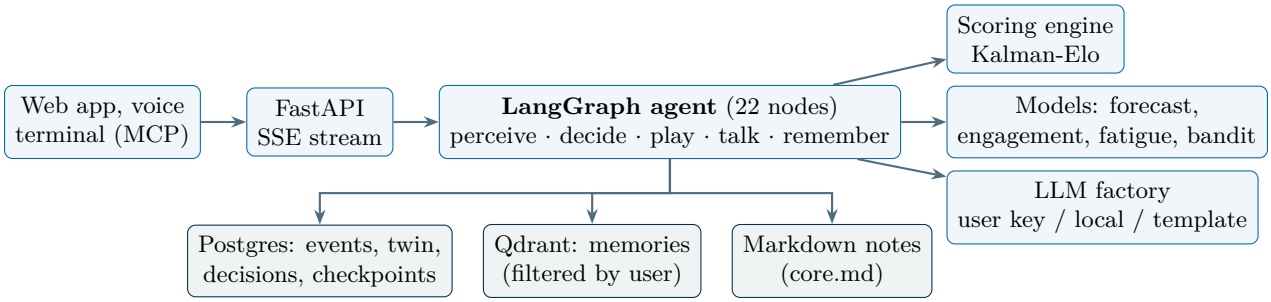


Figure 1: System overview. Postgres is the system of record; Qdrant holds searchable memories; Markdown holds the notes Lodestar writes.

Why this architecture. A graph agent (LangGraph 1.2) rather than a free-form tool loop makes every step inspectable and resumable: state is checkpointed in Postgres, so a browser crash in the middle of a game resumes on the same trial. Deterministic code decides *what* to do (which skill, which game, which difficulty) and records the decision with its candidates and the parts of each score; the language model only decides *how to say it*, from a facts packet. This split keeps decisions explainable and testable, keeps every number traceable, and lets the app run with a small local model or none at all. Users bring their own key for eight hosted providers, or use a local Ollama model; keys are encrypted at rest and decrypted only inside the LLM factory. All storage is Postgres (structured data, append-only events), Qdrant (memory search, always filtered by user) and readable Markdown notes.

The graph. Every turn enters at `ingest`, runs five perception nodes in parallel (load the twin, recall memories, read affect and context, forecast), then a `safety_gate` that refuses diagnosis requests before anything else happens, and `route_intent`, which reads the turn as play, conversation or a question to the Twin. The play path is `plan_next` → `prepare_exercise` → `play` (an interrupt: the graph waits for the browser to run the game) → `score_update` → `monitor` (continue, ease off, break or stop) → `reflect` → `coach`. The conversation path can call tools (set a goal, start an experiment, change a setting); any write first goes through `confirm`, an interrupt that shows the change and waits for a yes. Every path ends in `remember` and, at the end of a session, `consolidate`. Each node streams its start, end and timing to the browser, which is what lights up the Agent page.

Memory. Postgres holds the event log, trials, rounds, the twin's state and daily history, goals, experiments, decisions and traces. Qdrant holds three collections: *episodic* moments, distilled *facts* with a validity period, and a shared *knowledge* collection of tips written by the team. Recall prefetches dense (bge-small) and sparse (BM25) candidates filtered by user, fuses them, rescores by relevance, importance and time decay, and diversifies with MMR. Consolidation merges near-duplicates, promotes repeated observations to facts, expires old moments and rewrites `core.md`, the short profile the person can read and edit.

Design decisions and their trade-offs.

4 Assessment and the Brain Twin

Skills and tasks. Sixteen skills are grouped into attention, self-control, memory, speed, reasoning and language, each tagged *established*, *moderate* or *experimental* by the strength of its evidence. Classic paradigms (flanker, Stroop, n-back, Corsi, go/no-go, task switching, digit span, mental rotation, Tower of London, matrices) are rebuilt from their published descriptions, alongside original games and a Detective story whose chapters are games. A registry in YAML describes every game (skills and loadings, scoring, difficulty knobs, practice trials) and drives the UI, the scorer and the agent alike. A five-minute warm-up battery measures the core skills first.

Scoring. Each skill d has a belief $\theta_d \sim \mathcal{N}(\mu, \sigma^2)$ on a logit scale, each level a difficulty b. For an answer with

Decision	Instead of	Why
Graph agent, decisions in code	A free LLM tool loop	Decisions must be explainable, testable and resumable; the LLM only phrases them.
Kalman-Elo per skill	Fixed tests, or IRT batch fits	Updates after every answer, carries its own uncertainty, grows uncertain with time away; cheap enough to run per trial.
Facts packet + verifier	Prompting the LLM to be careful	Only checking can guarantee every number is real; a 4B local model passes after repair.
Postgres + Qdrant + Markdown	One vector database, or files	An append-only event log makes the profile rebuildable and auditable; search stays per user; notes stay readable.
Bring your own key, local fallback	One hosted provider	No cost to run, no data to a provider without consent, works offline.
Behaviour first for affect	Camera-first emotion AI	Works for everyone, needs no biometrics; the camera only adds a weak cue and never touches the profile.
Simulated norms, labelled	Borrowed norms on another scale	Showing a percentile we can't back would overstate accuracy; a linking study comes first.

Family	Skill	Evidence	Example games
Attention	Selective attention	established	Arrow Storm (flanker), Look Here (Posner), Odd One Out
	Sustained attention	established	Night Guard (vigilance)
	Divided attention	moderate	Control Tower, Two Screens (dual task)
Self-control	Inhibition	established	Ink Clash (Stroop), Hold It (go/no-go), Brake (stop-signal)
	Cognitive flexibility	established	Switch, Zigzag (trail switching), Secret Rules (card sort)
Memory	Working memory	established	Echo (n-back), Digits Back, Market Rush
	Short-term memory	established	Seen It?, Market List, Name Tags, Callback (delayed)
Speed	Processing speed	established	Quick Draw, Symbol Code, Twins, Connect
	Rhythm and timing	experimental	Off Beat
Reasoning	Reasoning	established	Pattern Gap (matrices), Logic Lock, Spot the AI's Mistake
	Spatial thinking	established	Turn It (rotation), Fold & Punch, Light Path (Corsi)
	Planning	moderate	Tower (Tower of London), Maze Escape, Guide the Robot
Language	Numerical thinking	established	Number Line, Quick Maths, Which Jar?
	Verbal skill	moderate	Word Burst (fluency), Scramble, Teach Me
Style	Decision style under risk	moderate	Balloon Pump (BART), Card Gamble (IGT)
	Self-awareness of accuracy	experimental	"How sure?" after about one answer in ten
	Creativity	experimental	Unusual Uses, rated against a rubric

Table 2: The 16 skills and one thinking style, with the strength of their evidence and the games that measure them. Styles are shown as styles, never as "better" or "worse".

outcome S (timed tasks use a high-speed high-stakes rule: right *and* within the reference pace),

$$p = \text{logistic}\Big(\tfrac{\mu - b}{\sqrt{1+\pi\sigma^2/8}}\Big), \qquad w = \text{loading} \times \text{reliability} \times \text{trust},$$
$$\sigma'^2 = \big(1/\sigma^2 + w\,p(1-p)\big)^{-1}, \qquad \mu' = \mu + \sigma'^2\, w\,(S - p).$$

Uncertainty grows with time away ($\sigma^2 \leftarrow \min(\sigma_0^2, \sigma^2 + q\,\Delta t)$) and with within-session process noise; a practice prior fitted to 1.08 million Lumosity learning curves moves μ by the expected gain per round. Cold-start priors come from skill correlations computed on 180,000 NCPT test runs. Estimates are shown as percentiles with 80% ranges ($\mu \pm 1.28\sigma$), labelled "simulated norms" until a linking study maps our scale onto published norms. Rounds that look like bad data (too fast, flat responding, outliers) are flagged and down-weighted rather than trusted.

History. A scored round is one transaction: events, trials, round, profile and daily history, under a per-user advisory lock. Events are never updated, so the whole profile can be rebuilt from the log. The Brain Twin shows each skill as a card (estimate, range bar, a shadow that widens with uncertainty), wired to skills of the same family and to the skills it correlates with most.

5 The AI coach and explainability

Each reply is built in three steps. The agent assembles a *facts packet* (numbered facts, decisions, knowledge tips and memories, each with an id); the LLM drafts in the chosen voice (Willow, Spark, Sage or Buddy); a verifier rejects any number not found in a cited fact, any banned clinical word and any unsupported claim, asks for a repair, and falls back to a template if the repair fails. In the UI every cited number shows its chip (`[F2]`), and "Why?" opens the decision behind a game: the candidate skills, the score of each, and its parts (uncertainty,

staleness, goal weight, bandit gain, fatigue, repetition). The Agent page shows the five stages lighting up live and every decision with the options it weighed; the full trace is stored per session. With the local `qwen3.5:4b` model, 3 of 4 first drafts passed verification and the fourth after a repair: 4 of 4 final replies were faithful, which is the point of the loop.

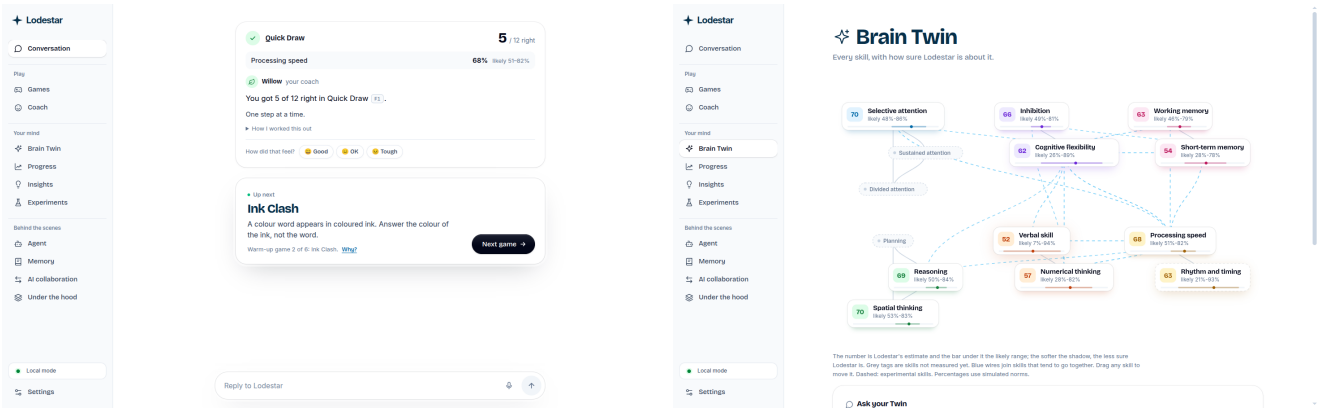


Figure 2: Left: after a game, the score, the skill it moved with its range, and the coach with a cited fact. Right: the Brain Twin, each skill a card with its estimate and range.

6 Adaptive learning and the dashboard

Difficulty targets the success rate the user chose (Chill, Balanced or Push): $b_{\text{next}} = \mu - \text{logit}(p^*)$, with an in-game staircase. On the simulator, Balanced mode achieved 77.7% success against an 80% target. **What to play** is chosen by a weighted score over skills (uncertainty, staleness, goals, experiments, fatigue, repetition) plus a Thompson-sampling bandit that learns which game moves each skill most for this person (+0.17 to +0.30 true-skill gain over random games, 95% interval). **Plans evolve**: goals are stated as "know this skill to within $\pm N$ points by a date", with rounds left estimated from the update rule; personal experiments compare morning and evening sessions and report a verdict only when $P(\text{one is better}) \geq 0.8$. The dashboard has the radar, progress curves with forecast bands and life-context strips, heatmaps by week and time of day, and goal cards; every estimate shows its range.

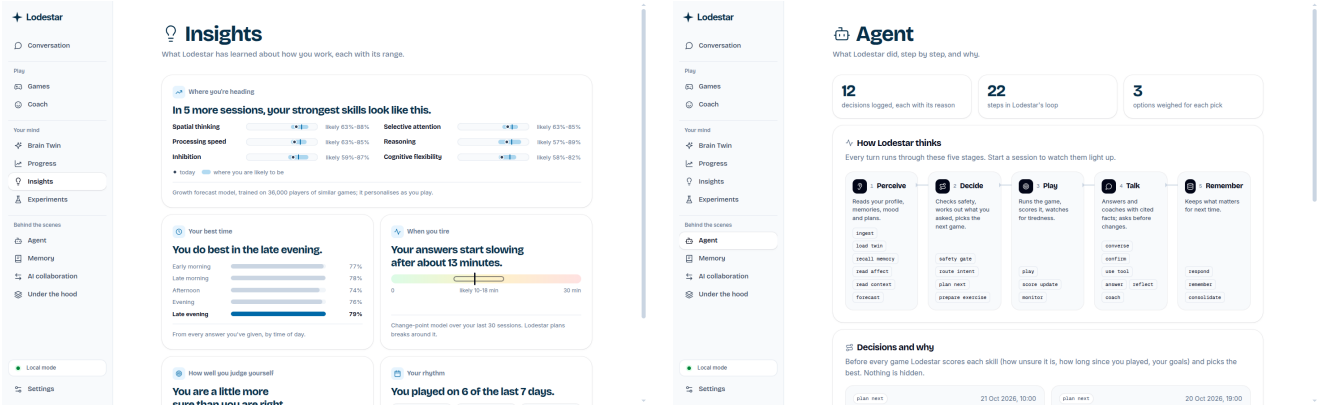


Figure 3: Left: Insights, each reading with its range and the model behind it. Right: the Agent page, the five stages and the decisions with their reasons.

Fresh content and games made to order. Items that could be memorised are generated fresh and pass five validators (schema, content rules, the banned-word filter, a solver that checks there is exactly one right answer, and novelty against what the person has already seen), with one repair round and a pre-filled pool so play never waits on a model. Matrices are generated by a program from rules, not by the LLM; for planted mistakes the program writes a correct solution, the LLM introduces one subtle error in 70% of items, and the solver re-runs the steps to confirm it. When a person asks for a new game ("remember a phone number while doing maths"), the LLM writes a recipe of known blocks (show items, distractor, recall, choice, rule, react); code validates it, a block engine runs it, and its scores count with reliability 0.4 until it has been tested more. The LLM never writes code that runs.

7 AI Collaboration Score

Each of five components is measured inside games: critical verification (finding planted mistakes in the AI's worked solutions, with consent), autonomy (solving before asking for a hint), iteration (a better retry after a failure), clarification (asking before acting on an ambiguous instruction) and responsible use (taking a suggested break). With a Beta(2,2) prior per component,

$$s_k = \frac{\text{successes}_k + 2}{\text{opportunities}_k + 4}, \qquad \text{ACS} = 100 \sum_k w_k s_k,$$

and an 80% range from 1,000 Monte Carlo draws. Weights are V 0.30, A 0.20, I 0.20, C 0.15, R 0.15: verification is raised from the brief's 0.25 because over-reliance is the main failure mode of AI assistance, clarification lowered because its opportunities are rarest. Ranks are stable under ±5-point weight changes (worst Spearman 0.997 on 200 simulated people).

8 Predictive models

- **Growth forecast.** LightGBM quantile models (10/50/90) with split-conformal calibration, trained on 36,297 users of Steyvers & Schafer's open Lumosity data and evaluated on held-out *users*: observed 80% interval coverage 79.6–80.0% for horizons of 1 to 20 sessions. In the app a person's own history outweighs the population forecast as data accumulate.
- **Engagement risk.** Logistic regression on eight behaviour features, trained on OULAD (25,273 students); held-out AUC 0.857 (LightGBM 0.863, days-since-last rule 0.828) with an inductive Venn–Abers range (ECE 0.006). It only ever triggers a shorter, easier session; the score is never shown.
- **Fatigue point.** A change-point detector on standardised log response times per session, combined with a Kaplan–Meier curve across sessions, gives the minute at which answers start slowing (median with its middle half). Sessions where the person slowed but became more accurate are excluded.
- **Best next game.** The Thompson-sampling bandit above, updated from every round.

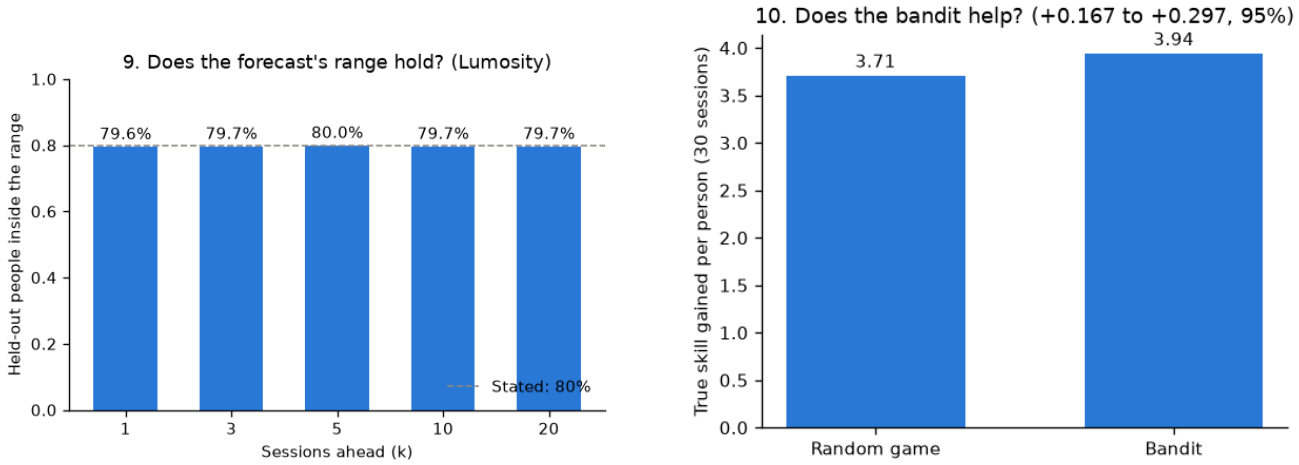


Figure 4: Left: the growth forecast's 80% ranges hold on held-out real players at every horizon. Right: true-skill gain with the bandit versus random games.

9 A session, end to end

1. The person says "I slept badly, something short". `route_intent` reads a quick session and records the life note; the twin and memories are loaded.
2. `plan_next` scores every skill, prefers short games, and records the decision with its candidates; the reason appears on the pick card with "Why?".
3. The game runs in the browser; each trial is sent as an event, so a crash resumes on the same trial.
4. `score_update` updates the skill in one transaction; `monitor` sees a slowdown past the person's fatigue point and offers a break instead of another round.
5. `reflect` builds the facts packet; `coach` drafts, verifies and repairs; the result card shows the score, the skill with its range, and the coach's cited sentence.

6. `remember` keeps the moment; at the end of the session `consolidate` updates the notes and facts the person can read on the Memory page.

10 Evaluation

A simulator draws 200 people with known true skills (correlated as in NCPT), plays ten sessions each through the scoring and selection code, and checks the profile against the truth. Table 3 summarises the twelve experiments; figures are in the repository. Simulated people are deliberately not ideal: they lapse, add trial noise, tire within a session, forget between sessions, learn at personal and per-game rates, and play on laggy devices.

Question	Result	Target
Recovery of true skill (correlation), after 1 / 3 / 10 sessions	0.773 / **0.820** / 0.905	> 0.8 at 3
80% ranges containing the true skill	**79.9%**	75–85%
Achieved success in Balanced mode	77.7%	$80 \pm 5\%$
Worst core-skill σ after the warm-up	0.65	< 0.8
Bad-data detection, precision / recall	1.00 / 1.00	> 0.9 / > 0.8
RMSE after 10 sessions: usual / long gaps / 50–150 ms lag	0.449 / 0.479 / 0.535	reported
ACS rank stability under weight shifts (Spearman)	0.997	> 0.9
Coach faithfulness, final replies (local 4B model)	4/4	100%
Forecast coverage on held-out real players	79.7%	80%
Bandit gain over random games (95% interval)	+0.17 to +0.30	reported
Sessions completed with all models down / Qdrant down / both	18/18 each	100%
Answers break the model: recovery after 3 / 10 sessions; ranges held	0.771 / 0.855; 71.0%	≥ 0.75; 70–90%

Table 3: Evaluation (`make eval`, seed 2026).

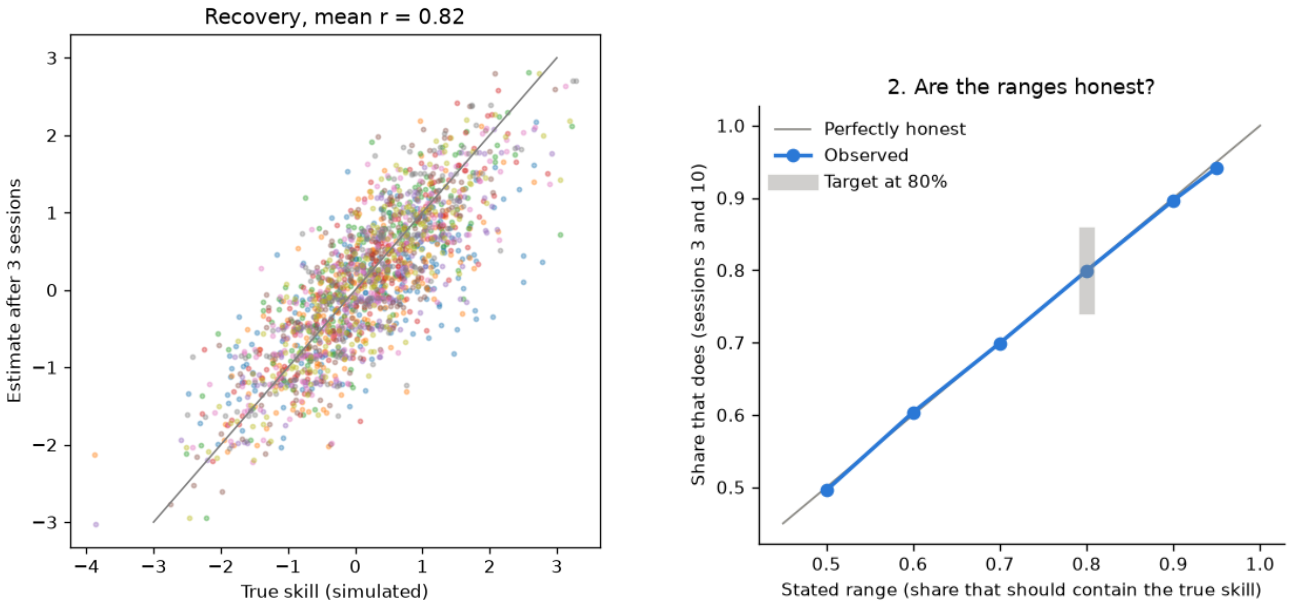


Figure 5: Left: recovery of true skill by session. Right: coverage of the 80% ranges.

Generalisation and robustness. Every model that touches people was trained on one set of real people and evaluated on others it never saw: 36,297 Lumosity players for the forecast (ranges held 79.7%), 25,273 OULAD students for engagement (AUC 0.857). To test the profile beyond its own assumptions, experiment 12 makes simulated people answer with a guessing floor, a personal discrimination and heavy-tailed noise while the scorer stays unchanged: recovery stays at $r = 0.771$ after three sessions (0.820 when answers follow the model) and 0.855 after ten; the 80% ranges hold 71% of the time, so they become somewhat overconfident rather than wrong. Experiments 1–5 also run through the full agent graph (`make sim-graph`), and the system was tested with sessions weeks apart, device lag, every model failing and the memory store unreachable. The software is covered by 265 backend tests, 84 frontend unit tests and 14 end-to-end suites in a real browser, run in CI.

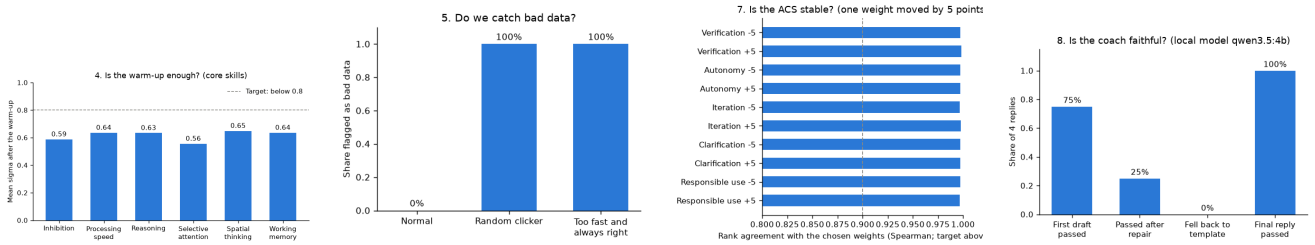


Figure 6: From left: uncertainty left after the five-minute warm-up; bad data caught; AI Collaboration Score rank stability under weight changes; coach faithfulness with a 4B local model.

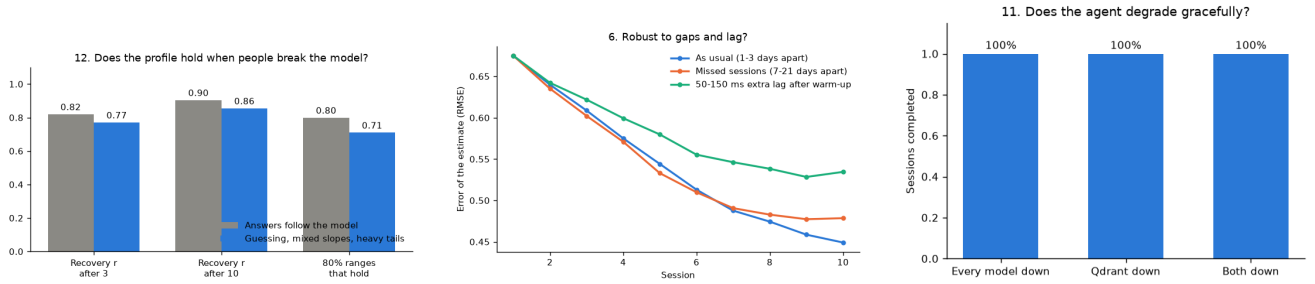


Figure 7: Generalisation and robustness. Left: recovery and coverage when answers break the scoring model. Middle: error after long gaps and with device lag. Right: sessions completed through the real agent with models, memory, or both unavailable.

11 Voice and emotion-aware coaching (bonus)

Voice. Tap to talk or say "Hey Lodestar" (Porcupine Web, with the person's own key and keyword file). Speech is transcribed by the browser, by Whisper on the server, or by Whisper running in the browser so audio never leaves the device; audio is discarded after transcription. Replies are spoken with one Kokoro voice per coach, citations are left out of speech, and every spoken line is on screen. Games can be voice-guided ("Ready. Go!") and some accept spoken answers.

Frustration and fatigue without a camera. The mandatory non-biometric path reads behaviour: error streaks, slowdowns against the person's own baseline, rising variability, rapid repeated taps, hint bursts, quitting and "too hard?". It can ease difficulty, switch to a lighter game, suggest a break or soften the coach's tone, and after it acts it asks "Was I right?" to collect labels. **Optional camera and voice tone.** With their own consent switches, MediaPipe face blendshapes (brow, eyes, mouth press) and speaking rate and pauses become one coarse "strain" number each, computed in the browser and sent alone; behaviour keeps at least 70% of the weight. A browser test watches every network call while the camera runs and allows exactly one coarse number to leave. These cues are described as strain cues, never as reading emotions.

12 Ethics, safety and data

Seven consent switches (core data, cloud AI, microphone, camera, voice tone, planted-mistake games, research export) are logged as events with the notice version; nothing is stored before the first-visit screens finish, and the app is for adults (18+). Camera and voice-tone cues are computed on the device, sent as one coarse number each, used only to suggest a break, and have no code path to the profile; no decision rests on face or voice alone. Users can read and forget any memory line, export everything, or delete everything and receive a receipt with the counts removed; a daily retention job removes old traces, checkpoints and events. Cloud AI receives only de-identified summaries. A banned-word list (diagnostic and clinical terms) is enforced in CI and in every reply, and a red-team suite checks requests for diagnoses. User API keys never appear in state, logs or exports.

13 Limitations and next steps

Validity rests on established paradigms and on the simulator, not yet on a study with real Lodestar users; percentiles use simulated norms until a linking study puts our scale against NCPT; the engagement model transfers from online courses to games; experimental skills (rhythm, creativity, accuracy awareness) are labelled as such. Next: a

consented pilot with 5–10 people (usability, whether "why" cards are understood, labels for the frustration cues), a linking study for norms, and more languages.

References

[1] S. Klinkenberg, M. Straatemeier, H. van der Maas. Computer adaptive practice of maths ability using a new item response model for on the fly ability and difficulty estimation. *Computers & Education*, 2011.

[2] Z. Buçinca, M. Malaya, K. Gajos. To trust or to think: cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *CSCW*, 2021.

[3] D. Jaffe et al. A large-scale normative dataset of cognitive performance (NCPT). *Scientific Data*, 2022.

[4] M. Steyvers, R. Schafer. Inferring latent learning factors in large-scale cognitive training data. *Nature Human Behaviour*, 2020.

[5] J. Kuzilek, M. Hlosta, Z. Zdrahal. Open University Learning Analytics dataset. *Scientific Data*, 2017.

[6] V. Vovk, I. Petej, V. Fedorova. Large-scale probabilistic predictors with and without guarantees of validity (Venn–Abers). *NeurIPS*, 2015.

[7] R. Baker et al. Towards sensor-free affect detection in cognitive tutor algebra. *EDM*, 2012.

[8] Google. MediaPipe Face Landmarker, 2023.